Wrap Up

In this project we have produced a number of models from Linear Modelling, Linear Modelling with transformed variables, Random Forests and Gradient Boosting Regression on Python to predict the average academic outcomes of students at schools based on various factors. We successfully implemented these models and compared them against a baseline score which we found at the start as an 'educated guess'. All of our models comfortably beat this baseline score, showing that these were at least somewhat able to predict the outcomes.

In the first section, we built the baseline model that we will then use throughout the project as a way of telling if our models were better than just guessing. We then produced code in Python to fit a linear model to our data using all of the available features, before trying to improve on these models with feature selection methods (specifically LASSO and PCA), and then trying to improve again by using transformations on the variables. Using these methods we were able to make big improvements on the baseline score of 8.91 and initial linear model we started with, which scored 3.88, significantly reducing the root mean squared error on the test data and the AIC when compared to the initial linear model. The best model we obtained in this section used a linear model fit to the full set of transformed variables based on our test RMSE results, which was able to achieve a test RMSE of 3.67. We also reused code we had previously found online that allowed us to analyse whether the fitted linear models were good ways of modelling the data. However, although we were able to make a large improvement over the baseline score, when examining both the initial model and the model with transformed variables, the plots suggested that using a linear model with the transformed variables we included is likely not the most effective way of modelling this data specifically, due to violations of the assumptions in a linear model. To improve on the limitations of these models, we could try to identify outliers and examine whether or not they should be excluded from our models, include more types of transformations on the variables and see if a linear model looks a better fit then, or even try to programme something that could identify which of the transformations looks as though it would best suit the relation to attainment8_average and just use that one, as opposed to adding every transformation and hoping to eliminate the irrelevant ones through feature selection.

In the second section, we used a Random Forest model, where we achieved a test RMSE of 3.879 and an R^2 of 0.810, showing that it was able to predict Attainment 8 scores reasonably accurately. The most important predictors were the average number of GCSE entries, school admission type and average qualification entries. The proportion of disadvantaged pupils was also important, suggesting that socioeconomic factors contributed to the model's predictions. We also investigated the effect of geography by removing local authority and parliamentary constituency, which slightly increased the cross-validation RMSE from 4.008 to 4.073.

One limitation we found was that the model performed considerably better on the training data than during cross-validation, suggesting some overfitting. We also compared ordinary K-Fold with GroupKFold, which keeps observations from the same school together. This increased RMSE from 4.008 to 4.305, suggesting that the model had more difficulty predicting results for schools it had not encountered before. Although Random Forest performed well overall, these results show that its accuracy depends partly on how the data is divided, and that the model may not generalise as well to entirely unseen schools.

Our final model section explored the Gradient Boosting model, which showed that school and pupil characteristics could be used to predict average Attainment 8 scores with reasonable accuracy. Starting with 10 predictors, we found that expanding the model to 23 predictors improved its performance, suggesting that including a wider range of school and pupil information was beneficial. Hyperparameter tuning improved the results further, while applying PCA did not lead to better predictions. The final model achieved a **test RMSE of 4.562**, showing that Gradient Boosting was able to capture useful relationships between school characteristics and educational attainment.

However, the model also had limitations. Cross-validation grouped by school and local authority produced higher prediction errors than ordinary cross-validation, suggesting that the model may perform less accurately on schools or areas it has not encountered before. This highlights the importance of considering how well a model generalises, rather than relying only on its test RMSE. Overall, Gradient Boosting was a useful approach to predicting attainment, although its performance should be considered alongside the other models and the different modelling decisions made across the group.

Comparing the final test RMSE values, transformed Linear Regression performed best at **3.67**, followed by Random Forest at **3.879** and Gradient Boosting at **4.562**. All three models significantly improved on our baseline RMSE of **8.91**, showing that more complex algorithms do not necessarily produce better predictions. However, the comparison was not entirely fair, as each model used different predictors and preprocessing methods. In particular, Gradient Boosting excluded examination-entry variables that were included in the other models, which may partly explain its higher RMSE. Furthermore, grouped cross-validation produced higher errors for Random Forest and Gradient Boosting, suggesting weaker performance on entirely unseen schools. Therefore, although transformed Linear Regression achieved the lowest test RMSE, we cannot conclude that it is inherently the best algorithm. A more reliable comparison would require consistent predictors and validation methods across all three models.

In summary, we were able to find information on and implement various data modelling methods. The results we got showed each of our models were able to predict academic outcomes for schools with relatively high degrees of accuracy when compared to a baseline score. The best model which we were able to produce based on the root mean squared error from predicting the

test data was the linear model produced with all available transformed variables with a score of 3.67. However, the plots showed that the linearity and normality assumptions of the linear model could have been violated, therefore these models likely could still be improved further to produce an even better model, perhaps by exploring the discussed limitations of the models.

Our models all agreed that the most significant factors when predicting average academic performance of students were the average number of GCSE's (or total qualifications) entered, the proportion of the students considered disadvantaged (or whose disadvantaged status was unknown) and the admission type of the school. The number of GCSE's entered and admission type of the school were both factors that could potentially be questioned as each factor has a plausible relation to academic performance. More GCSE entries would give pupils more chances to get higher scores. Similarly, if a school admitted students on academic grounds, those students would likely perform better. Nevertheless, we considered both factors worth retaining, as they provide an insight into differences in school expectations and admissions policies. In particular, the number of qualifications entered may show how much schools encourage or provide opportunities for pupils to pursue a broader range of qualifications. The disadvantaged pupil proportion, on the other hand, is a particularly interesting finding, which illustrates a potential link between socioeconomic background and academic performance.